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Design and Implementation of a Voice over Wi-Fi System Using SIP and RTP Protocols

Preliminary Technical Report for the Term Project

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PROJECT DEFINITION AND THEORETICAL FOUNDATION:

1. Scientific Problem:

Traditional telephone systems often depend on dedicated telephony infrastructure and specialized switching equipment. Where a wireless local area network is already available, deploying a separate telephone network for internal communication may introduce additional installation, maintenance, and management requirements.

The scientific and engineering problem addressed in this project is how to design and implement a centralized IP-based telephone system that provides internal voice communication to users connected through Wi-Fi, while allowing expansion with supplementary PBX services and evaluation using objective network-performance measurements.

The project investigates the integration of a PBX platform, SIP-based endpoint registration and call control, and real-time voice transport over a local wireless IP network. The scope is local Wi-Fi-based VoIP; it is not carrier-grade cellular Wi-Fi Calling based on IMS/ePDG architecture.

2. Project Aim and Objectives:

The main aim is to design, deploy, and test a Wi-Fi-based VoIP communication system using FreePBX as the IP-PBX platform and Zoiper as the softphone client.

- Study the principles and architecture of VoIP and wireless local area networks.
- Deploy FreePBX in a virtualized environment and configure its initial operation.
- Create SIP extensions using PJSIP technology.
- Register Zoiper clients with the PBX server and establish internal voice calls.
- Configure supplementary services, including Call Recording and Voicemail, and extend the system with further PBX features.
- Prepare a repeatable test environment for future QoS evaluation using delay, jitter, packet loss, throughput, and suitable voice-quality indicators.
- Explore a future AI-based approach for predicting call-quality degradation from measured network-performance data.

3. State of the Art:

Research on VoIP over wireless networks has examined signaling protocols, network behavior, and metrics used to evaluate perceived voice quality. The following studies provide a basis for the proposed work.

3.1 Performance Evaluation of VoIP Protocols in Wireless Networks

Noori et al. compare SIP and H.323 over an IEEE 802.11a wireless network using the QualNet simulator. Their evaluation considers session traffic, establishment time, RTP traffic, end-to-end delay, and throughput. The study illustrates how signaling choices and wireless-network conditions can be investigated using measurable performance indicators [1].

3.2 Performance Evaluation of Voice over Wi-Fi Using IMS

Aziz examines Voice over Wi-Fi in an IP Multimedia Subsystem (IMS) context and discusses factors such as codec selection, Wi-Fi technology, user movement, distance, data rate, and packet loss. This operator-oriented architecture differs from the local FreePBX-based PBX architecture in this project; it is relevant as related work on wireless voice performance, but the architectures are not identical [2].

3.3 VoIP Quality Metrics

Adhilaksono and Setiawan review VoIP quality metrics, including jitter, bandwidth, packet loss, and throughput. They also discuss Quality of Experience (QoE) measures such as Mean Opinion Score (MOS), Perceptual Evaluation of Speech Quality (PESQ), and Peak Signal-to-Noise Ratio (PSNR). Their review supports selecting measurable indicators for a later performance-evaluation stage [3].

3.4 Position of the Proposed Project:

The present project emphasizes a working local PBX environment using FreePBX/Asterisk and Zoiper, verification of extension registration and internal calling, and the addition of supplementary PBX services. The next research step is controlled measurement under varying wireless conditions. AI-based prediction is a future extension, not a completed result.

Aspect	Related research	Present project
Environment	Wireless simulation or IMS-based VoWiFi	Local Wi-Fi LAN with FreePBX/Asterisk
Focus	Protocol comparison and quality metrics	Practical PBX deployment and internal calling
Evaluation	Simulation or performance analysis	Initial functional tests; quantitative QoS testing planned
Further direction	Voice quality and network performance	Controlled QoS study and possible AI-based prediction

4. Proposed Solution:

The proposed solution is a centralized Wi-Fi-based VoIP system. FreePBX provides the web-based PBX administration platform, while Asterisk performs the underlying call-processing functions. PJSIP is used for SIP endpoint configuration. Zoiper applications act as user telephone clients connected to the same local IP network through Wi-Fi.

A user registers a Zoiper account using an assigned extension number and authentication credentials. When a call is placed, SIP signaling is handled by the PBX to locate and alert the destination extension. After the session is established, RTP carries real-time audio between the communication endpoints. The PBX can additionally provide services such as voicemail and call recording.

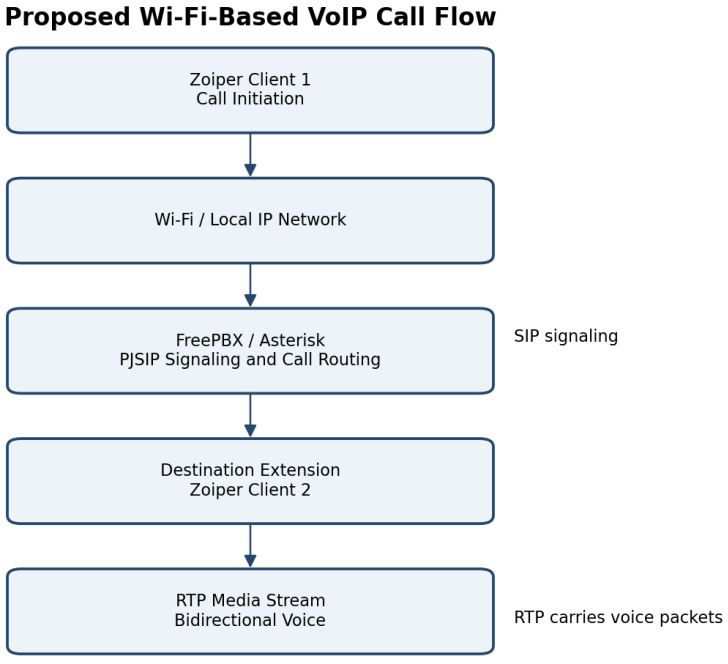


Figure 1. Proposed Wi-Fi-Based VoIP Call Flow

4.1 Main System Components:

Component	Function
FreePBX	Web-based administration and configuration of the PBX.
Asterisk	Core telephony engine responsible for call processing.
PJSIP	SIP endpoint and signaling functionality used for extensions.
Zoiper	Softphone application installed on user devices.
Wi-Fi access network	Wireless IP connectivity between clients and server.
VMware virtual machine	Hosts the FreePBX server environment.

4.2 Current Status and Planned Work:

Completed and demonstrated: FreePBX 17 installation in VMware, access to the administration interface, creation of PJSIP extensions, Zoiper client configuration, and a successful internal call between two clients. An initial audio issue was resolved by granting microphone permission to Zoiper.

In progress or planned: validation of supplementary services, including Call Recording and Voicemail; implementation of Ring Groups, IVR, and CDR; and subsequent QoS measurements. AI-based call-quality prediction is a proposed future research direction.

CHAPTER ONE: THEORETICAL STUDY

1.1 Introduction:

This chapter presents the theoretical concepts required to understand the design and implementation of the proposed Wi-Fi-based VoIP system. It introduces telephone switching systems, IP-PBX architecture, VoIP communication, wireless LANs, signaling and media protocols, codecs, and network-quality measures.

1.2 Traditional Telephone Systems and PBX:

A traditional telephone network commonly uses circuit switching, in which a communication path is reserved for a call for its duration. Within an organization, a Private Branch Exchange (PBX) manages calls between internal extensions and may provide access to external telephone lines. Typical functions include extension dialing, call transfer, call forwarding, voicemail, and call routing.

1.3 IP-PBX:

An IP-PBX performs telephone switching and call management over IP networks. Users may connect through wired Ethernet or wireless LANs. IP-PBX systems integrate voice services with data networks and allow software clients and IP phones to use centrally managed extensions.

1.4 Voice over Internet Protocol:

VoIP carries voice communication over packet-switched IP networks. A microphone captures the speaker's voice, which is sampled and encoded by a codec. The encoded data is packetized and transmitted through the network. At the receiving endpoint, packets are processed, decoded, and played through a speaker. VoIP separates call-control signaling from the transport of voice media.

1.5 Wi-Fi and WLAN:

Wi-Fi is a family of wireless local-area networking technologies based on IEEE 802.11 standards. A wireless access point allows client devices to join a local network without a physical Ethernet connection. In this project, Wi-Fi is the access technology used by softphone clients to reach the local IP-PBX. Performance can be affected by signal strength, interference, contention, distance, channel utilization, and background traffic.

1.6 IP-PBX Architecture: FreePBX and Asterisk:

Asterisk is an open-source telephony engine that provides call-processing and media-related capabilities. FreePBX is a web-based administration framework used to configure and manage an Asterisk-based PBX. In this project, FreePBX provides the administrative interface, while Asterisk performs the underlying telephony functions. Extensions, routing rules, and supplementary services are configured through FreePBX.

1.7 Session Initiation Protocol (SIP):

SIP is an application-layer signaling protocol used to create, modify, and terminate multimedia sessions. It does not normally carry voice audio itself. Common SIP methods include REGISTER, INVITE, ACK, BYE, CANCEL, and OPTIONS. REGISTER allows a client to register its contact information; INVITE initiates a session; ACK confirms the final response to an INVITE; and BYE terminates an established session.

1.8 PJSIP:

PJSIP is an open-source SIP protocol stack and multimedia communication library. Asterisk uses PJSIP to provide SIP endpoint functionality. In FreePBX, PJSIP extensions are configured with endpoint and authentication settings. A Zoiper client uses the corresponding account details to register with the PBX.

1.9 RTP and RTCP:

RTP is commonly used to transport real-time audio and video. RTP packets include sequence numbers and timestamps that help the receiver reconstruct the media stream and identify packet ordering and timing. RTP itself does not guarantee delivery, bandwidth, or low delay. RTCP is used alongside RTP to exchange control and quality-related information, such as reception statistics.

1.10 Audio Codecs:

An audio codec encodes and decodes voice samples for transmission. Codec choice affects bit rate, bandwidth consumption, processing requirements, and potential audio quality. Examples include G.711, G.729, G.722, and Opus. The codecs actually negotiated depend on the capabilities and configuration of both endpoints and the PBX.

Codec	**General characteristics**
G.711	Pulse Code Modulation; commonly uses a 64 kbit/s audio payload rate and is widely supported.
G.729	Low-bit-rate speech codec designed to reduce bandwidth requirements; licensing and implementation considerations may apply.
G.722	Wideband speech codec providing a wider speech frequency range than narrowband telephony.
Opus	Flexible interactive audio codec supporting a wide range of bit rates and audio applications.

1.11 Quality of Service and Quality of Experience:

Quality of Service (QoS) describes measurable network-performance characteristics. Quality of Experience (QoE) describes the quality perceived by the end user. Packet-level measurements help explain network behavior, while user-oriented or objective audio-quality measures help assess the resulting listening experience.

Parameter	**Meaning and relevance to VoIP**
Delay / Latency	Time taken for packets or voice information to travel between endpoints; excessive delay disrupts conversation.
Jitter	Variation in packet arrival times; high jitter can cause uneven playback and require buffering.
Packet loss	Packets that fail to reach the destination; loss can produce gaps or distortion in audio.
Throughput	Successfully delivered data rate; available capacity must support voice traffic and other network use.
MOS	A commonly used 1-to-5 rating scale for perceived speech quality; the method used to obtain MOS must be stated.

A future evaluation should define the measurement method, test duration, network conditions, number of concurrent calls, and codec settings. Measurements should be repeated under controlled conditions for fair comparison.

1.12 Virtualization and Test Environment:

Virtualization allows a software-defined computer, called a virtual machine, to run on a physical host. VMware Workstation is used in this project to host the FreePBX server. The virtual machine's network mode and IP configuration must permit communication with the Zoiper clients. Bridged, NAT, and host-only networking have different connectivity implications; the selected mode should match the intended test topology.

1.13 Chapter Summary:

This chapter introduced the theoretical foundations of the project, including IP-PBX systems, VoIP, Wi-Fi, SIP/PJSIP signaling, RTP media transport, codecs, QoS/QoE, supplementary PBX services, and virtualization. These concepts support the design decisions and practical configuration of the project.

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